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(54) **SYSTEM AND METHOD FOR GENERATING A VIRTUAL EXPERIENCE IN A VIRTUAL ENVIRONMENT**

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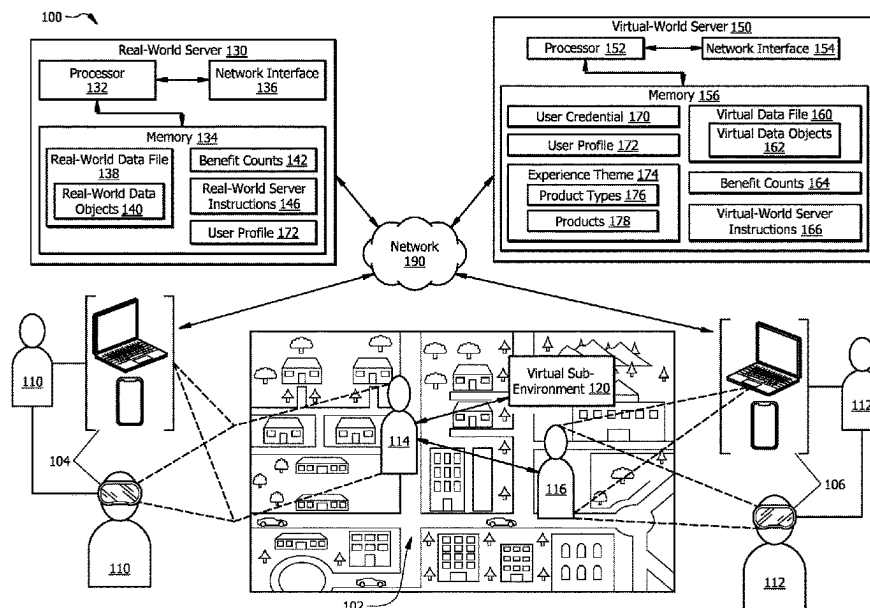
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(57) **ABSTRACT**

A system includes a memory and a processor coupled to the memory. The processor receives a first user credential associated with a first user and authorizes an avatar of the first user to enter a virtual environment. The processor receives a first selection of an experience theme and a second selection of at least one product type from the selected experience theme. The processor determines a first data file that provides a first benefit count to the first user when a first amount of data objects is transferred from a first data file of the first user to a second data file. The processor indicates to the first user that the first data file provides the first benefit count to the first user, receives a command to obtain the first data file, and initiates a registration process to register the first user to obtain the first data file.

20 Claims, 3 Drawing Sheets



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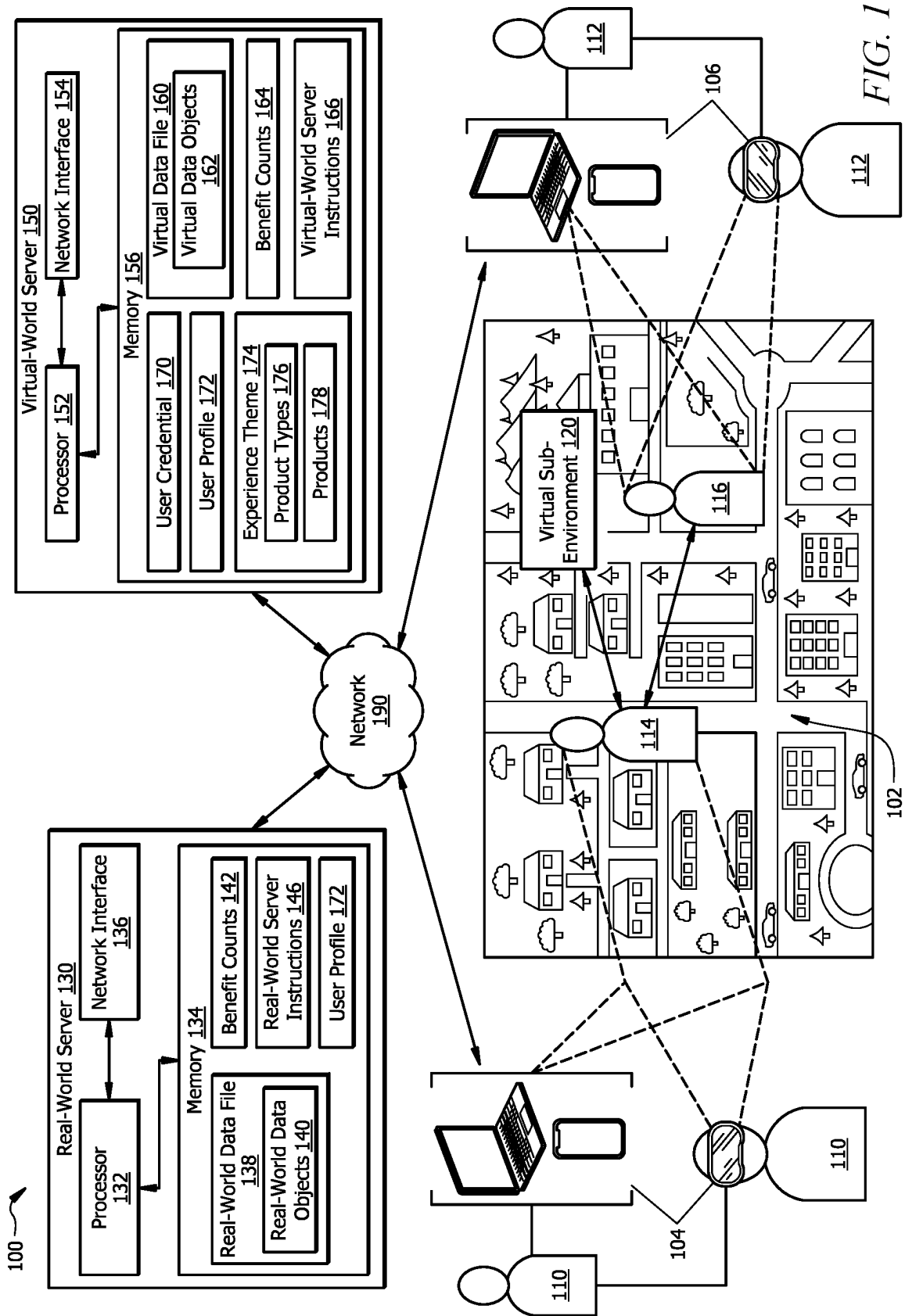
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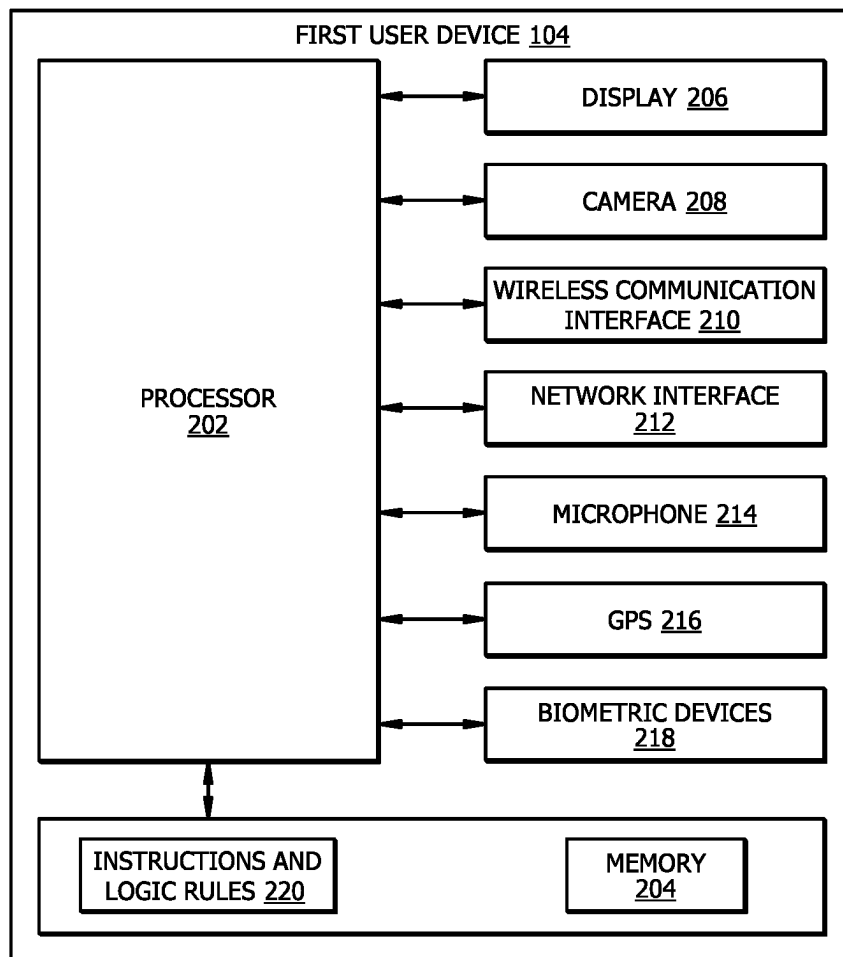


FIG. 2

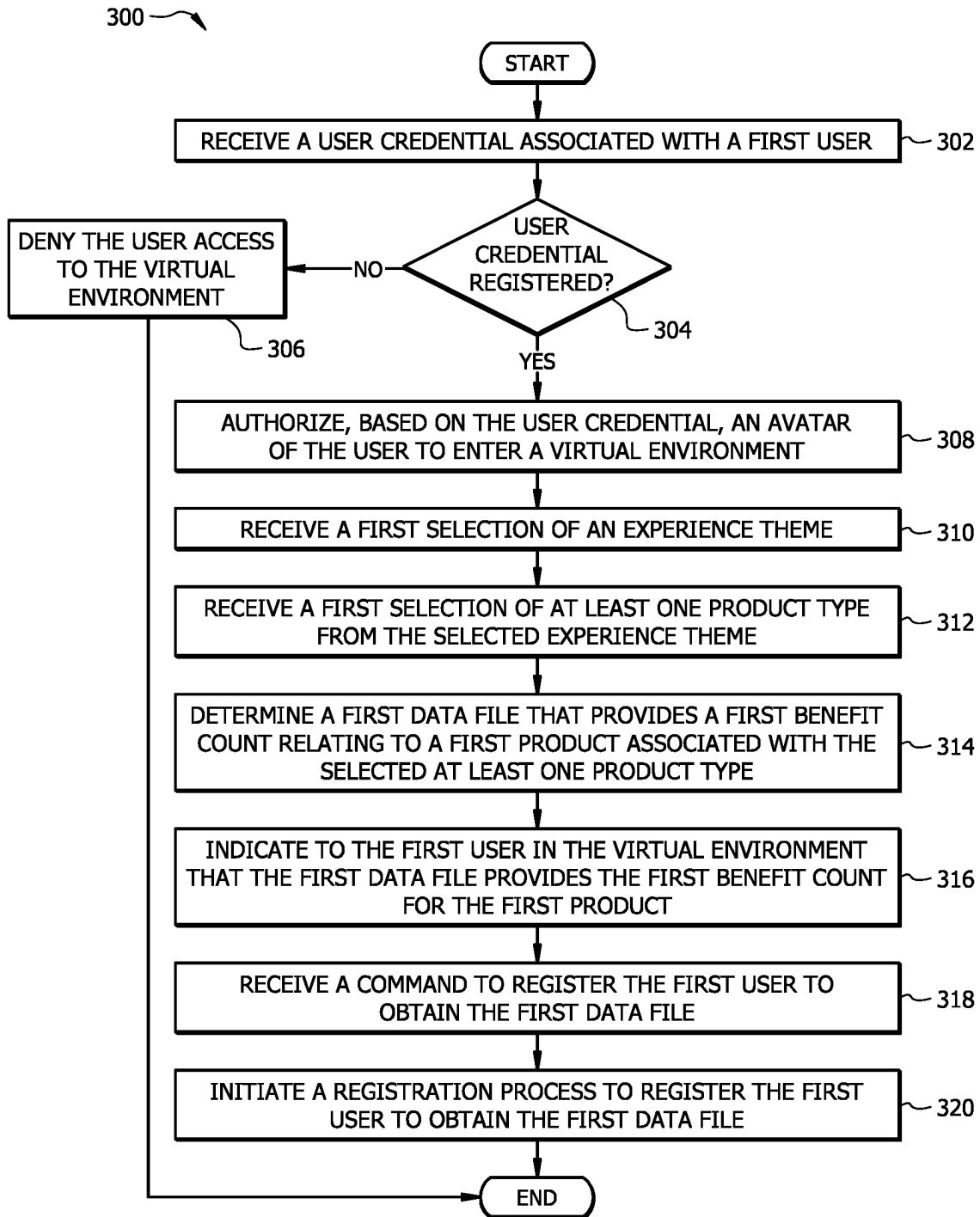


FIG. 3

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SYSTEM AND METHOD FOR GENERATING A VIRTUAL EXPERIENCE IN A VIRTUAL ENVIRONMENT

TECHNICAL FIELD

The present disclosure relates generally to network communication, and more specifically to a system and method for generating a virtual experience in a virtual environment.

BACKGROUND

In a network environment, user devices are in data communication with other user devices that may be distributed anywhere in the world. These network environments allow data and information to be shared among these devices. Some of the technical challenges that occur when data is exchanged between devices are controlling data leakage, unauthorized access to data, and preventing malicious activities. Data storing user devices, such as computers, laptops, augmented reality devices, virtual reality devices, and smartphones, are vulnerable to attacks. This vulnerability poses several network security challenges. Existing systems are typically unable to detect a malicious attack until after the attack has occurred. For example, a bad actor may pretend to be another user in a virtual environment which then allows the bad actor to gain access to other users' information.

SUMMARY

The system and methods implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by allowing a user to securely access a virtual environment and perform secure data interactions in the virtual environment. The disclosed system and methods provide several practical applications and technical advantages.

For example, the disclosed system and methods provide the practical application of improving interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in a real-world environment may be used in a virtual environment (e.g., metaverse environment) to authenticate the user before allowing the user to access the virtual environment and perform any kind of action or interaction within the virtual environment. Additionally or alternatively, as described in embodiments of the present disclosure, user information collected from the user and/or assigned to the user in the real-world environment may be used in the virtual environment (e.g., metaverse environment) to provide access to products within the virtual environment. This process provides improved information security because it authenticates that an avatar is associated with the user and not an unauthorized party and that the user is authorized to access the virtual environment, and products provided in the virtual environment.

Thus, the disclosed system and methods improve data security in the virtual environment. By improving data security in virtual environment, the disclosed system and methods generally improve technology related to performing secure data interactions in a virtual environment.

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The disclosed system and methods provide the additional practical application of saving memory resources. The seamless data flow between the real-world systems and virtual-world systems as a result of interoperability of these systems allows each system to store less data by avoiding the need to store the same data (e.g., authentication data, login credentials, etc.) in both systems, as data stored in one system can be accessed, or otherwise leveraged, by the other system. This saves memory resources by avoiding duplication of data. The saving of memory resources may leave more system memory for storing other data used in other technical operations. This provides the additional technical advantage of improving processing efficiency of computing systems that manage the real-world and virtual word environments.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

FIG. 1 is a schematic diagram of a system, in accordance with certain aspects of the present disclosure;

FIG. 2 is a block diagram of an embodiment of the first user device used by the system of FIG. 1; and

FIG. 3 illustrates a flowchart of an example method for performing data interactions in a virtual environment, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

Example System

FIG. 1 is a schematic diagram of a system 100, in accordance with certain aspects of the present disclosure. System 100 may include a first user device 104, a second user device 106, real-world server 130, and a virtual-world server 150, each connected to a network 180. A first user 110 is associated with the first user device 104 and a second user 112 is associated with the second user device 106. The system 100 may be communicatively coupled to the communication network 190 and may be operable to transmit data between each one of the first user device 104, second user device 106, real-world server 130, and virtual-world server 150 through the communication network 190.

In general, the system 100 may improve interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, access to virtual sub-environments and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in a real-world environment may be used in a virtual environment 102 (e.g., metaverse environment) to authenticate the first user 110 before allowing the first user 110 to access the virtual environment 102 and perform any kind of action or interaction within the virtual environment 102. Additionally or alternatively, user information collected from the first user 110 and/or assigned to the first user 110 in the real-world environment or virtual environment 102 may be used in the virtual environment 102 (e.g., metaverse environment) to provide the first user 110 access to products and/or services within the virtual environment 102. This process provides improved information security because it authenticates that

a first avatar **114** is associated with the first user **110**, not an unauthorized party, and that the first user **110** is authorized to access the virtual environment **102**, obtain products and services within the virtual environment **102** and perform data interactions. Further, a user may obtain a product by transferring real-world data objects between real-world entities based on data interactions performed in the virtual environment **102**.

It may be noted that the terms “real-world” and “real-world environment” in this disclosure refer to any non-virtual environment where users (e.g., users **110** and **112**) can physically interact with real persons and objects. A real-world data interaction may refer to any data interaction performed outside the virtual environment **102** (e.g., a metaverse environment). Further, it may be noted that while certain embodiments of the present disclosure may be described in the context of a metaverse environment which is an example of a virtual environment **102**, the methods discussed in this disclosure apply to any other virtual environment **102**. The terms “virtual environment” and “metaverse environment” are used interchangeably throughout this disclosure. Furthermore, it may be noted that while certain embodiments of this disclosure describe one or more operations in relation to the first user **110**, these embodiments apply to any other user (e.g., second user **112**) connected to network **190**.

The first user **110** may access the virtual environment **102** (e.g., a metaverse environment) through the first user device **104**. The first user device **104** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of the virtual environment **102** to the first user **110**. Examples of a virtual environment **102** may include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. The virtual environment **102** may be configured to use realistic or non-realistic physics for the motion of objects within the virtual environment **102**. For example, some virtual environments **102** may be configured to use gravity whereas other virtual environments **102** may not be configured to use gravity. Within the virtual environment **102**, each user may be associated with an avatar (such as the first avatar **114** for the first user **110**). An avatar is a graphical representation of a user at a virtual location within the virtual environment **102**. In embodiments, the virtual location of the avatar may be correlated to the physical location of a user in the real-world environment. Examples of an avatar may include, but are not limited to, a person, an animal, or an object. In some embodiments, the features and characteristics of the avatar may be customizable, and user defined. For example, the size, shape, color, attire, accessories, or any other suitable type of appearance features may be specified by a user. By using an avatar, a user is able to move within the virtual environment **102** to interact with one or more avatars and objects within the virtual environment **102** while independently remaining at a physical location in the real-world environment or being in transit in the real-world environment.

While engaging in the virtual environment **102** via the first avatar **114**, the first user **110** may interact with a plurality of other users, objects and/or entities through a respective avatar. For example, the second user **112** may attempt to engage in an interaction session with the first avatar **114** through a second avatar **116** associated with the second user **112**. In another example, the first avatar **114** of the first user **110** may access a virtual sub-environment **120** within the virtual environment **102** and perform virtual data

interactions within the virtual sub-environment **120**. In the real-world environment, the second user **112** may be physically located at a distance away from the first user **110**. The second user **112** may access the virtual environment **102** through the second user device **106** to control the second avatar **116** and attempt to engage in an interaction session with the first user **110** through the first avatar **114**.

Before the interaction between the first avatar **114** and the second avatar **116** occurs or the first avatar **114** can access the virtual sub-environment **120**, the virtual-world server **150** may authenticate that the first avatar **114** is associated with the first user **110** and not an unauthorized third-party. For example, the first user **110** may be required to sign into a secure portal that provides access to a data file (e.g., real-world data file **138** and/or virtual data file **160**) associated with the first user **110**. As shown in FIG. 1, a real-world data file **138** of the first user **110** is stored and managed by the real-world server **130** and a virtual data file **160** of the first user **110** is stored and managed by the virtual-world server **150**. In one or more embodiments, the virtual-world server **150** may employ single sign-on (SSO), multifactor authentication, or any other suitable authentication scheme in order to allow the first user **110** access to the virtual data file **160**. Similarly, the real-world server **130** may employ single sign-on (SSO), multifactor authentication, or any other suitable authentication scheme in order to allow the first user **110** access to the real-world data file **138**. The virtual data file **160** and the real-world data file **138** may include virtual data objects **162** and real-world data objects **140** respectively owned by the first user **110**. The real-world server **130** and the virtual-world server **150** may store other information related to the first user **110** including, but not limited to, user profile information, account information (e.g., including identity and other details relating to the respective data files **138** and **160**), avatar information, digital assets (e.g., respective real-world data objects **140** and virtual data objects **162**) information, or any other suitable type of information that is associated with a user within the virtual environment **102** and/or the real-world environment.

As shown in FIG. 1, virtual-world server **150** comprises a processor **152**, a memory **156**, and a network interface **154**. The processor **152** comprises one or more processors operably coupled to the memory **156**. The processor **152** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **152** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **152** is communicatively coupled to and in signal communication with the memory **156**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **152** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **152** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., virtual-world server instructions **166**) to implement the virtual-

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world server **150**. In this way, processor **152** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the virtual-world server **150** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The virtual-world server **150** is configured to operate as described with reference to FIGS. 1 and 3. For example, the processor **152** may be configured to perform at least a portion of the method **300** as described in FIG. 3.

The memory **156** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **156** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

The memory **156** is operable to store user credential **170**, user profile **172**, experience themes **174**, product types **176**, products **178**, virtual data file **160**, virtual data objects **162**, benefit counts **164** and the virtual-world server instructions **166**. The virtual-world server instructions **166** may include any suitable set of instructions, logic, rules, or code operable to execute the virtual-world server **150**.

The network interface **154** is configured to enable wired and/or wireless communications. The network interface **154** is configured to communicate data between the virtual-world server **150** and other devices, systems, or domains (e.g. user devices **104** and **106** and real-world server **130**). For example, the network interface **154** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **152** is configured to send and receive data using the network interface **154**. The network interface **154** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

The real-world server **130** comprises a processor **132**, a memory **134**, and a network interface **136**. The processor **132** comprises one or more processors operably coupled to the memory **134**. The processor **132** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **132** may be a programmable logic device, a microcontroller, a micro-processor, or any suitable combination of the preceding. The processor **132** is communicatively coupled to and in signal communication with the memory **134**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **132** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **132** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., real-world server instructions **146**) to implement the real-world server **130**. In this way, processor **132** may be a special-purpose computer designed to implement the functions disclosed

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herein. In one or more embodiments, the real-world server **130** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The real-world server **130** is configured to operate as described with reference to FIGS. 1 and 3. For example, the processor **132** may be configured to perform at least a portion of the method **300** as described in FIG. 3.

The memory **134** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **134** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

The memory **134** is operable to store information relating to real-world data file **138**, real-world data objects **140**, benefit counts **142** and the real-world server instructions **146**. The real-world server instructions **146** may include any suitable set of instructions, logic, rules, or code operable to execute the real-world server **130**.

The network interface **136** is configured to enable wired and/or wireless communications. The network interface **136** is configured to communicate data between the real-world server **130** and other devices, systems, or domains (e.g. user devices **104** and **106**, and virtual-world server **150**). For example, the network interface **136** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **132** is configured to send and receive data using the network interface **136**. The network interface **136** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

Each of the real-world server **130** and the virtual-world server **150** is generally a suitable server (e.g., including a physical server and/or virtual server) operable to store data in a memory (e.g., respective memories **134** and **156**) and/or provide access to application(s) or other services. One or both of the real-world server **130** and the virtual-world server **150** may be a backend server associated with a particular entity (e.g., organization) that facilitates conducting data interactions between entities, between one or more users, and/or between a user and an entity. In other embodiments, one or both of the real-world server **130** and the virtual-world server **150** may be organized in a distributed manner, or by leveraging cloud computing technologies. Real-world server **130** may store information which is primarily used to support data interactions performed in the real-world environment. Virtual-world server **150** may store information which is primarily used to support data interactions performed in the virtual environment **102** (e.g., a metaverse environment). It may be noted that the operations performed by the real-world server **130** and the virtual-world server **150** described in embodiments of the present disclosure may be implemented by a single server.

The communication network **190** may facilitate communication within the system **100**. This disclosure contemplates the communication network **190** being any suitable network operable to facilitate communication between the first user device **104**, second user device **106**, real-world server **130**, and virtual-world server **150**. Communication network **190** may include any interconnecting system capable of transmitting audio, video, signals, data, messages, or any combination of the preceding. Communication network **190** may include all or a portion of a local area

network (LAN), a wide area network (WAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, etc.), a Long Term Evolution (LTE) network, a Universal Mobile Telecommunications System (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a Near Field Communication network, a Zigbee network, and/or any other suitable network, operable to facilitate communication between the components of system 100. In other embodiments, system 100 may not have all of these components and/or may have other elements instead of, or in addition to, those above.

Each of the user devices (i.e., first user device 104 and second user device 106) may be any computing device configured to communicate with other devices, such as a server (e.g., real-world server 130, virtual-world server 150), databases, etc. through the communication network 190. Each of the user devices may be configured to perform specific functions described herein and interact with one or both of real-world server 130 and the virtual-world server 150, e.g., via respective user interfaces. Each of the user devices is a hardware device that is generally configured to provide hardware and software resources to a user. Examples of a user device include, but are not limited to, a virtual reality device, an augmented reality device, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, or any other suitable type of device. The user devices may comprise a graphical user interface (e.g., a display), a touchscreen, a touchpad, keys, buttons, a mouse, or any other suitable type of hardware that allows a user to view data and/or to provide inputs into the user device. Each user device may be configured to allow a user to send requests to one or both of real-world server 130 and the virtual-world server 150, or to another user device. Example User Device

FIG. 2 is a block diagram of an embodiment of the first user device 104 used by the system of FIG. 1. First user device 104 may be configured to display the virtual environment 102 (referring to FIG. 1) within a field of view of the first user 110 (referring to FIG. 1), capture biometric, sensory, and/or physical information of the first user 110 wearing the first user device 104, and to facilitate an electronic interaction between the first user 110 and the second user 112 (referring to FIG. 1) or between the first user 110 and an entity (e.g., represented by a virtual entity in the virtual environment 102).

First user device 104 comprises a processor 202, a memory 204, and a display 206. Further embodiments may include a camera 208, a wireless communication interface 210, a network interface 212, a microphone 214, a global position system (GPS) sensor 216, and/or one or more biometric devices 218. First user device 104 may be configured as shown or in any other suitable configuration. For example, first user device 104 may comprise one or more additional components and/or one or more shown components may be omitted.

The processor 202 comprises one or more processors operably coupled to and in signal communication with memory 204, display 206, camera 208, wireless communication interface 210, network interface 212, microphone 214, GPS sensor 216, and biometric devices 218. Processor 202 is configured to receive and transmit electrical signals among one or more of memory 204, display 206, camera 208, wireless communication interface 210, network inter-

face 212, microphone 214, GPS sensor 216, and biometric devices 218. The electrical signals are used to send and receive data (e.g., images captured from camera 208, virtual objects to display on display 206, etc.) and/or to control or communicate with other devices. Processor 202 may be operably coupled to one or more other devices (for example, the real-world server 130 and/or virtual-world server 150 shown in FIG. 1).

The processor 202 is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor 202 may be a programmable logic device, a microcontroller, a micro-processor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor 202 may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor 202 may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions to implement the function disclosed herein, such as some or all of those described with respect to FIGS. 1 and 3. For example, processor 202 may be configured to display virtual objects on display 206, detect hand gestures, identify virtual objects selected by a detected hand gesture, capture biometric information of a user, such as first user 110, via one or more of camera 208, microphone 214, and/or biometric devices 218, and communicate via wireless communication interface 210 with the real-world server 130, virtual-world server 150 and/or second user device 106. In some embodiments, the function described herein is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

The memory 204 is operable to store any of the information described with respect to FIGS. 1 and 3 along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor 202. For example, the memory 204 may store the instructions 220. The memory 204 comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory 204 is operable to store, for example, information relating to the identity of the user (e.g., at least a portion of user profile 172), instructions for performing the functions of first user device 104 described herein, and any other data or instructions. The memory 204 may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

Display 206 is configured to present visual information to a user (for example, first user 110 in FIG. 1) in a virtual reality environment, an augmented reality environment or mixed reality environment. In other embodiments, the display 206 is configured to present visual information to the user as the virtual environment 102 (referring to FIG. 1) in

real-time. In an embodiment, display **206** is a wearable optical display (e.g., glasses or a headset) configured to reflect projected images and enables a user to see through the display. For example, display **206** may comprise display units, lens, semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units include, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an active-matrix OLED (AMOLED), an organic LED (OLED) display, a projector display, or any other suitable type of display as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. In another embodiment, display **206** is a graphical display on a user device. For example, the graphical display may be the display of a tablet or smart phone configured to display virtual environment **102**.

Examples of camera **208** include, but are not limited to, charge-coupled device (CCD) cameras and complementary metal-oxide semiconductor (CMOS) cameras. Camera **208** is configured to capture images of a wearer of first user device **104**, such as first user **110**. Camera **208** may be configured to capture images continuously, at predetermined intervals, or on-demand. For example, camera **208** may be configured to receive a command from first user **110** to capture an image. In another example, camera **208** is configured to continuously capture images to form a video stream. Camera **208** is communicably coupled to processor **202**.

Examples of wireless communication interface **210** include, but are not limited to, a Bluetooth interface, an RFID interface, a near field communication interface, a local area network (LAN) interface, a personal area network interface, a wide area network (WAN) interface, a Wi-Fi interface, a ZigBee interface, or any other suitable wireless communication interface as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. Wireless communication interface **210** is configured to facilitate processor **202** in communicating with other devices. For example, wireless communication interface **210** is configured to enable processor **202** to send and receive signals with other devices, such as second user device **106**, the real-world server **130** and/or virtual-world server **150** (referring to FIG. **1**). Wireless communication interface **210** is configured to employ any suitable communication protocol.

The network interface **212** is configured to enable wired and/or wireless communications. The network interface **212** is configured to communicate data between the first user device **104** and other network devices, systems, or domain(s). For example, the network interface **212** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **202** is configured to send and receive data using the network interface **212**. The network interface **212** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

Microphone **214** is configured to capture audio signals (e.g., voice signals or commands) from a user, such as first user **110**. Microphone **214** is configured to capture audio signals continuously, at predetermined intervals, or on-demand. Microphone **214** is communicably coupled to processor **202**.

GPS sensor **216** is configured to capture and to provide geographical location information. For example, GPS sensor **216** is configured to provide a geographic location of a user, such as first user **110**, employing first user device **104**. GPS

sensor **216** may be configured to provide the geographic location information as a relative geographic location or an absolute geographic location. GPS sensor **216** may provide the geographic location information using geographic coordinates (i.e., longitude and latitude) or any other suitable coordinate system. GPS sensor **216** is communicably coupled to processor **202**.

Examples of biometric devices **218** may include, but are not limited to, retina scanners, fingerprint scanners and facial scanners. Biometric devices **218** are configured to capture information about a person's physical characteristics and to output a biometric signal based on captured information. A biometric signal is a signal that is uniquely linked to a person based on their physical characteristics. For example, biometric device **218** may be configured to perform a retinal scan of the user's eye and to generate a biometric signal for the user based on the retinal scan. As another example, a biometric device **218** is configured to perform a fingerprint scan of the user's finger and to generate a biometric signal for the user based on the fingerprint scan. Biometric device **218** is communicably coupled to processor **202**.

Referring back to FIG. **1**, in one or more embodiments, one or both of the real-world server **130** and the virtual-world server **150**, and one or more user devices (e.g., second user device **106**) may be part of an Information Technology (IT) infrastructure of an entity or organization. For example, second user **112** may be a representative of the entity or organization who may use the second user device **106** to enter the virtual environment **102** and virtually interact with one or more users (e.g., first user **110**) via the second avatar **116** to provide services to the first user **110**.

The real-world server **130** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the real-world environment. Similarly, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the virtual-world server **150** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). When the real-world server **130** and/or the virtual-world server **150** are owned and/or operated by a particular entity or organization (e.g., are part of the IT infrastructure of the entity or organization), being registered with the real-world server **130** and/or the virtual-world server **150** may also be interpreted as being registered with the particular entity or organization. For example, when the first user **110** is registered with the virtual-world server **150**, this may be interpreted as the first user **110** being registered with the entity or organization that owns and/or manages the virtual world server **150**. In one embodiment, the real-world server **130** and the virtual-world server **150** are owned and/or operated by the same entity/organization. In this context, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). In alternative embodiments, the real-world server **130** and the virtual-world server **150** may be owned and/or operated by different entities/organizations. For example, the real-world server **130** may be owned and/or operated by a first entity and the virtual-world server **150** may be owned and/or operated by a second entity different from the first entity.

In one or more embodiments, as the first user **110** initially registers with the real-world server **130** in the real-world environment, the real-world server **130** may collect several pieces of information from the user including information

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relating to the identity of the user such as legal name, social security number, biometrics (e.g., fingerprints, retina scans, face ID etc.), residence address, phone numbers, assets owned by the user, and copies of government issued documents (e.g., drivers permit, state identity card etc.). This information is stored by real-world server 130 as part of user profile 172 of the first user 110. In one embodiment, at least a portion of the user profile 172 relating to the first user 110 collected in the real-world environment may be stored in the virtual-world server 150. Once the identity of the first user 110 is confirmed and all other information provided by the first user 110 is verified to be correct, the real-world server 130 may generate a real-world data file 138 for the first user 110 in which the first user 110 may store real-world data objects 140 owned by the first user 110. In one example, the first user 110 may engage in a real-world interaction with a service representative managing the real-world server 130 (e.g., physical interaction at an office location, over phone, voice chat etc.) to provide such information that can be used to register the first user 110 at the real-world server 130 and generate the real-world data file 138 of the first user 110. In another example, the first user 110 may engage in a real-world interaction by accessing a webpage provided and managed by the real-world server 130. Once the first user 110 initiates a registration process via the webpage, the real-world server 130 may walk the first user 110 through several steps in which the first user 110 may be requested to provide information necessary to verify the identity of the first user 110 and register the first user 110 with the real-world server 130.

Information relating to the real-world data file 138 of the first user 110 may be stored as part of the user profile 172 of the first user 110. This information may include, but is not limited to, an identity (e.g., unique name or number) of the real-world data file 138, amount of real-world data objects 140 stored in the real-world data file 138, a log of data interactions conducted in relation to the real-world data file 138 and any other information relating to the real-world data file 138. As described below, the information relating to the real-world data file 138 of the first user 110 may further include a benefit count 142 associated with the real-world data file 138.

Once registered with the real-world server 130, the real-world server 130 may allow the first user 110 to perform one or more data interactions in the real-world environment. For example, a real-world data interaction may include transferring one or more real-world data objects 140 from the real-world data file 138 of the first user 110 to a second real-world data file (not shown) of the second user 112 or another entity. Another example data interaction may include receiving one or more real-world data objects 140 in the real-world data file 138 of the first user 110 from the second real-world data file of the second user 112 or another entity. Another example data interaction may include requesting by the first user 110 transfer of real-world data objects from a data file of a second user to a user data file of a third user as part of satisfying an agreement between the first user 110 and the third user. Another example data interaction may include modifying at least a portion of the user profile 172 (e.g., user credentials to access the real-world server, phone numbers, residential address, email address, information relating to user assets etc.) stored at the real-world server 130. It may be noted that a data interaction in accordance with embodiments of the present disclosure refers to any interaction in the real-world environment and/or virtual environment 102 that includes transfer of data

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between computing nodes (e.g., first user device 104, second user device 106, real-world server 130 and virtual-world server 150).

The first user 110 may additionally register with the virtual-world server 150. In one embodiment, when initially registering with the virtual-world server 150, the first user 110 may provide to the virtual-world server 150 a credential (e.g., username and password) that provides the first user 110 access to the real-world server 130. In one embodiment, a single web page or web portal may allow the first user 110 to register with the real-world server 130 as well as the virtual-world server 150. The first user 110 may first register with the real-world server 130 as described above and generate credentials that allow the first user 110 access to the real-world server 130 and services provided by the real-world server 130. Once registered with the real-world server 130, the web portal may provide the first user 110 an option to additionally register with the virtual-world server 150 which may allow the first user 110 to perform data interactions in the virtual environment 102. Registration with the virtual-world server 150 may include generating a user credential 170 that allows the first user 110 to log in to the virtual-world server 150 and enter the virtual environment 102 via first avatar 114 of the first user 110. Once registered with the virtual-world server 150, the first user 110 may generate a virtual data file 160 in which the first user 110 may store virtual data objects 162 owned by the first user 110. In one or more embodiments, the virtual data file 160 of the first user 110 is associated with the real-world data file 138 of the first user 110. For example, the virtual data file 160 is a virtual image of the real-world data file 138, wherein the virtual data objects 162 correspond to the real-world data objects 140. In other words, the virtual data file 160 is a virtual representation of the real-world data file 138. In another example, the virtual data file 160 stores a portion of the real-world data objects 140 in the form of virtual data objects 162. In another example, real-world data objects 140 may be converted to virtual data objects 162, and vice versa. In this case, there may not be a one-to-one conversion between the real-world data objects 140 and virtual data objects 162. For example, one real-world data object 140 may be converted to a plurality of virtual data objects 162, wherein the conversion ratio may dynamically change from time to time. In one embodiment, when the real-world server 130 and the virtual-world server 150 are owned and/or managed by the same entity or organization, the first user 110 may use the same user credential 170 to login to the real-world server 130 as well as the virtual-world server 150.

Information relating to the virtual data file 160 of the first user 110 may be stored by the virtual-world server 150 as part of the user profile 172 of the first user stored at the virtual-world server 150. This information may include, but is not limited to, an identity of the virtual data file 160, amount of virtual data objects 162 stored in the virtual data file 160, a log of virtual data interactions conducted in the virtual environment 102 in relation to the virtual data file 160 and any other information relating to the virtual data file 160. As described below, the information relating to the virtual data file 160 of the first user 110 may further include a benefit count 164 associated with the virtual data file 160.

In certain embodiments, when the real-world server 130 and the virtual-world server 150 are owned and/or managed by different entities, first user 110 may separately register with one of the real-world server 130 and the virtual-world server 150 without registering with the other. For example, when the real-world server 130 is registered with a first entity and the virtual-world server 150 is registered with a

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second entity different from the first entity, first user **110** may separately register with the virtual-world server **150** without registering or needing to register with the real-world server **130**. For example, as the first user **110** separately registers with the virtual-world server **150**, the virtual-world server **150** may collect several pieces of information from the first user **110** including information relating to the identity of the user such as legal name, social security number, biometrics (e.g., fingerprints, retina scans, face ID etc.), residence address, phone numbers, assets owned by the user, and copies of government issued documents (e.g., drivers permit, state identity card etc.). This information is stored by virtual-world server **150** as part of user profile **172** of the first user **110**. In one embodiment, at least a portion of the user profile **172** relating to the first user **110** collected in by the virtual-world server **150** may be stored in the real-world server **130**. Once the identity of the first user **110** is confirmed and all other information provided by the first user **110** is verified to be correct, the virtual-world server **150** may generate the virtual data file **160** for the first user **110** in which the first user **110** may store virtual data objects **162** owned by the first user **110**.

Once registered with the virtual-world server **150**, the virtual-world server **150** may allow the first user **110** to perform one or more virtual data interactions. For example, a virtual data interaction may include transferring one or more virtual data objects **162** from the virtual data file **160** of the first user **110** to a second virtual data file (not shown) of the second user **112** or another entity. Another example data interaction may include receiving one or more virtual data objects **162** in the virtual data file **160** of the first user **110** from the second virtual data file of the second user **112** or another entity. Another example data interaction may include requesting by the first user **110** transfer of virtual data objects **162** from a data file of a second user to a data file of a third user as part of satisfying an agreement between the first user **110** and the third user. Another example data interaction may include modifying at least a portion of the user profile **172** (e.g., user credentials to access the real-world server, phone numbers, residential address, email address, information relating to user assets etc.) stored at the virtual-world server **150**.

In one or more embodiments, the virtual data file (e.g. virtual data file **160**) is a software application running on a computing node owned and/or operated by the respective user (e.g., first user **110**). For example, when the first user **110** desires to receive virtual data objects **162** from a virtual data file of the second user **112**, first user **110** may direct the second user **112** to a unique cryptographic address (e.g., public key) issued by the virtual data file **160**. In one embodiment, the virtual data file **160** may not itself store the virtual data objects **162** but may store information that points to a location of the virtual data objects **162**, for example, on a server (e.g., virtual-world server **150**). Virtual data file **160** may be web-based or hardware-based. For example, virtual data file **160** may be stored in a mobile device or a desktop computer connected to the internet. Additionally or alternatively, virtual data file **160** may be stored in a device (e.g., USB drive) that is not connected to the network **190**.

In certain embodiments, virtual environment **102** may include one or more virtual sub-environments **120**. A virtual sub-environment **120** may be a designated region within the virtual environment **102** for use by a user (e.g., first user **110**) or an entity/organization. For example, a virtual sub environment **120** may be assigned to a particular entity which may provide one or more services to users of the virtual

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environment **102**. The virtual sub-environment **120** may take any form in the virtual environment **102** including, but not limited to, a virtual room, a virtual building or portions thereof, a virtual store, or any other designated virtual space/region within the virtual environment **102**. Virtual-world server **150** may be configured to generate virtual sub-environments **120** with the virtual environment **102**. A virtual sub-environment **120** is usually generated for a customized purpose. For example, a virtual sub-environment **120** may be used by an entity that is assigned the virtual sub-environment **120** for receiving virtual data objects **162** from users for storing in virtual data files **160** and/or for distribution to specified one or more target entities. Further a virtual sub-environment may allow users (e.g., first user **110**) of the virtual environment **102** to perform virtual data interactions including transferring virtual data objects **162** to other users (e.g., second user **112**) and/or one or more target entities. As described below, virtual-world server **150** may be configured to generate customized virtual sub-environments **120** that may be used to provide customized virtual experiences to users.

Data security is important in any system that supports online data interactions between computing nodes of the system. Online data interactions in real-world environments have existed for several decades and robust measures are already in place to ensure data security in real-world systems. However, virtual-world technology (e.g., metaverse technology) is relatively new and data security is a challenge in metaverse systems as the development of metaverse related technologies is still at a nascent stage and standardized systems that provide robust data security are not yet in place.

Embodiments of the present disclosure leverage user data collected and/or assigned in a real-world environment for providing data security in a metaverse environment. In general, embodiments of the present disclosure describe techniques that improve interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security and other data interactions performed in real-world and virtual environments. In one or more embodiments, the system and methods discussed herein use information stored in a user profile **172** of the first user **110** generated during real-world data interactions with the first user **110** to allow the first user **110** to securely obtain products and/or services in the virtual environment **102**.

In one or more embodiments, virtual-world server **150** may be configured to use a user credential **170** collected from the first user **110**, generated by the first user **110** or assigned to the first user **110** during real-world data interactions with the first user **110**, to verify identity of the first user **110** in the virtual environment **102**. Thus, the user credential **170** provides the first user **110** access to the virtual environment **102**. For example, the user credential **170** may be used by the virtual-world server **150** to verify that the first avatar **114** belongs to and is controlled by the first user **110**.

In one example, user credential **170** may include a username and password generated by the first user **110** as part of registering with the real-world server **130**. The virtual-world server **150** may allow the first user **110** to use the same username and password to enter the virtual environment **102** via first avatar **114**. In an alternative embodiment, the first user **110** may generate a separate user credential (different from the user credential **170**) as part of registering with the real-world server **130** and use the separate user credential to log into the real-world server **130** and perform real-world

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data interaction. The first user 110 may separately generate user credential 170 as part of registering with the virtual-world server 150 (e.g., during a real-world data interaction with the real-world server 130). Once the first user 110 has registered with the virtual-world server 150 and generated the user credential 170, the first user 110 may use the user credential 170 to log into the virtual-world server 150 and perform virtual-world data interactions. The user credential 170 of the first user 110 may be associated with the user profile 172 of the first user 110.

In one embodiment, a second user 112 may gain access to the virtual-world server 150 and enter the virtual environment 102 without registering with the virtual-world server 150 and/or real-world server 130. In other words, virtual-world server 150 may allow the second user 112 to enter the virtual environment 102 as a guest user. For example, in order to enter the virtual environment 102 as a guest user, second user 112 may need to complete a simple verification process that may include verifying the identity of the second user 112 (e.g., using email verification, text verification etc.) and setting up a temporary avatar (e.g., second avatar 116) to navigate the virtual environment 102. In one embodiment, if the identity of the second user 112 is successfully verified, virtual world server 150 may generate a temporary user credential 170 which the second user 112 may use to access the virtual environment 102 as a guest user. It may be noted that, a guest user who is not registered with the real-world server 130 or virtual-world server 150 may not have access to the full array of products and/or services provided by the respective real-world server 130 or virtual-world server 150.

Virtual-world server 150 may be configured to facilitate a user (e.g., first user 110) to obtain one or more products in the virtual environment 102 or real-world environment. As shown in FIG. 1, virtual-world server 150 may be configured to store one or more experience themes 174, wherein each experience theme 174 relates to a particular type of experience that can be had by the user 110 in the virtual environment 102 or the real-world environment. An example of an experience theme associated with the virtual environment 102 may include visual elements that make up the virtual environment 102. An example of an experience theme associated with the real-world environment may include a travel experience. An experience theme 174 includes one or more product types 176, wherein each product type 176 is a type of product 178 that relates to the experience theme 174. For example, when the experience theme 174 relates to visual experience of the first user 110 within the virtual environment 102, example product types 176 associated with this experience theme 174 may include, but are not limited to, avatars, buildings, roads, objects, colors and patterns. In another example, when the experience theme 174 relates to a travel experience in the real-world environment, product types 176 associated with this experience theme 174 may include, but are not limited to, cars, buses, trains, travel destinations, hotels, restaurants, and excursions. An experience theme 174 may include several products 178 associated with each product type 176. For example, when the experience theme 174 relates to a visual experience within the virtual environment 102, the “avatars” product type 176 may include several specific avatars 114 that may be used as the first avatar 114 of the first user 110 within the virtual environment 102. The “buildings” product type 176 may include a plurality of specific virtual buildings that can be used to generate the virtual environment 102. The “colors” and “patterns” product type 176 may include a plurality of colors and patterns respectively that can be used to render specific objects or portions

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of the virtual environment 102. When the experience theme 174 relates to a travel experience in the real-world environment, the “destination” product type 176 may include a selection of different destinations, the “cars” product type 176 may include a selection of different cars that be obtained for a travel, the “hotels” product type may include a selection of hotels at the selected destination and the like.

In operation, when the first user 110 enters the virtual environment (e.g., using first avatar 114) after logging in to the virtual-world server 150 by providing user credential 170, the first user 110 may navigate to a virtual sub-environment 120 that is configured to facilitate users to obtain products 178 associated with one or more experience themes 174. The virtual sub-environment 120 may be assigned to a particular real-world entity. Once the first user 110 has entered the virtual sub-environment 120, the virtual-world server 150 may present to the first user 110 one or more experience themes 174. For example, the virtual-world server 150 may display the experience themes 174 on a virtual display within the virtual sub-environment 120. Virtual-world server 150 may be configured to allow the first user 110 to select one of the presented experience themes 174. The first user 110 may select one of the experience themes 174 depending on what kind of experience the first user 110 is looking for. For example, the virtual-world server 150 may present a visual experience theme associated with the virtual environment 102 and a travel experience theme associated with the real-world environment. In this example, the first user 110 selects the visual experience theme associated with the virtual environment 102. Certain embodiments of the present disclosure will now be described in accordance with an example in which the first user 110 selects the visual experience theme as an experience theme 174. However, it may be noted that these embodiments apply in relation to other experience themes 174.

Once the virtual-world server 150 receives a selection of an experience theme 174, virtual-world server 150 may be configured to present to the first user 110 within the virtual sub-environment 120, one or more product types 176 associated with the selected experience theme 174. For example, when the first user 110 selects the visual experience theme as the selected experience theme 174, virtual-world server 150 presents a list of product types 176 associated with the visual experience theme including, but not limited to, avatars, buildings, roads, objects, colors and patterns. Virtual-world server 150 may be configured to allow the first user 110 to select one or more of the presented product types 176 associated with the selected experience theme 174. As the first user 110 selects each product type 176, virtual-world server 150 may be configured to access from the memory 156 one or more products 178 associated with the selected product type 176 and present the product 178 to the first user 110 in the virtual environment 102. For example, when the first user 110 selects the “avatars” product type 177, virtual-world server 150 may access from the memory 156 one or more specific avatars associated with the “avatars” product type 176 and present the one or more specific avatars to the first user 110 in the virtual-sub environment 120. For example, virtual-world server 150 may display a 2-dimensional or 3-dimensional image of one or more avatars associated with the avatar product type 176. Virtual-world server 150 may allow the first user 110 to select at least one product 178 (e.g., specific avatar) associated with a selected product type 176. In one embodiment, virtual-world server 150 may allow the first user 110 to select multiple product types 176 related to the selected experience theme 174 and one or more products 178 associated with each of the

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multiple selected product types 176. For example, the first user 110 may want to build a customized virtual sub-environment 120 and may want to obtain specific visual elements (e.g., avatars, buildings, roads, objects, colors, patterns etc.) to build the virtual sub-environment 120. For this purpose, the first user 110 may select a specific avatar from the “avatar” product type 176, one or more specific buildings from the “buildings” product type 176, one or more roads from the “roads” product type 176, one or more colors from the “colors” product type 176, and one or more patterns from the “patterns” product type 176.

Once the product selections have been made by the first user 110, virtual-world server 150 may be configured to determine one or more data files (e.g., real-world data files 138 or virtual data files 160) that can be used to obtain at least one of the products 178 selected by the first user 110. In this context, virtual-world server 150 may be configured to store information relating to a plurality of real-world data files 138 and virtual data files 160 that may be used by the first user 110 to obtain products 178 associated with experience themes 174. A particular product 178 may be obtained by the first user 110 by transferring a pre-configured amount of real-world data objects 140 from a real-world data file 138 registered to the first user 110 to another real-world data file 138 registered to an entity that provides the particular product 178, and/or by transferring a pre-configured amount of virtual data objects 162 from a virtual data file 160 registered to the first user 110 to another virtual data file 160 registered to the entity that provides the particular product 178. Different products 178 may be obtained by transferring different pre-configured amounts of real-world data objects 140 or virtual data objects 162 to respective data files of entities that provide the products 178. For example, a first product 178 may be obtained by the first user 110 by transferring a first pre-configured amount of real-world data objects 140 or virtual data objects 162, and a second product 178 may be obtained by the first user 110 by transferring a second pre-configured amount of real-world data objects 140 or virtual data objects 162.

In some embodiments, a single data file (e.g., real-world data file 138 or virtual data file 160) may be used to obtain multiple products 178. In some embodiments, any one of multiple data files (e.g., multiple real-world data file 138, multiple virtual data files 160 or combinations thereof) may be used to obtain a same product 178.

In one or more embodiments, a data file (e.g., real-world data file 138 or virtual data file 160) may provide a benefit count 142/164 to the first user 110, when the first user 110 uses the data file (e.g., real-world data file 138 or virtual data file 160) to obtain a particular product 178 (e.g., by transferring data objects to a respective data file of an entity that provides the particular product 178). For example, the first user 110 may receive a pre-configured benefit count 142 when the first user 110 uses a particular real-world data file 138 to obtain a particular product 178. Similarly, the first user 110 may receive a pre-configured benefit count 164 when the first user 110 uses a particular virtual data file 160 to obtain a particular product 178. In one embodiment, different data files (e.g., different real-world data files 138 and/or different virtual data files 160) may provide different benefit counts 142/164 to obtain a same product 178. In an additional or alternative embodiment, a single data file (e.g., real-world data file 138 or virtual data file 160) may provide different benefit counts 132/164 to obtain different products 178.

In one or more embodiments, benefit count 142 associated with a real-world data file 138 may be used to obtain one or

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more products 178 in the real-world environment, and benefit count 164 associated with a virtual data file 160 may be used to obtain one or more products 178 in the virtual environment 102. A higher benefit count 142/164 generally allows the first user 110 to obtain more products 178. Thus, a higher benefit count 142/164 is generally of higher value. The first user 110 may accumulate benefit counts 142/164 received as a result of using a particular data file (e.g., real-world data file 138 or virtual data file 160) to obtain several products 178.

The information stored by the virtual-world server 150 in memory 156 relating to a plurality of real-world data files 138 and virtual data files 160 that may be used to obtain products 178 associated with experience themes 174 may include one or more of an identity of each real-world data file 138 or virtual data file 160, products 178 that may be obtained using the real-world data file 138 or the virtual data file 160, and benefit counts 142 or 164 provided by the real-world data file 138 or virtual data file 160 respectively when used to obtain each of the products 178.

As described above, once the selection of the products 178 associated with a selected experience theme 174 has been made by the first user 110, virtual-world server 150 may be configured to determine one or more data files (e.g., real-world data files 138 or virtual data files 160) that can be used to obtain one or more of the products 178 selected by the first user 110. For a selected product 178, virtual-world server 150 determines one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files that the first user 110 may receive when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the product 178. For example, based on the selected product 178, virtual-world server 150 may be configured to search the memory 156 (e.g., information related to data files stored in the memory 156) for data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178. Based on the search, virtual-world server 150 may determine one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files that the first user 110 may receive when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the product 178.

The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) that the determined one or more data files (e.g., real-world data file 138 and/or virtual data file 160) may provide to the first user 110 the corresponding benefit counts 142/164 when the determined one or more data files are used to obtain the product 178. For example, virtual-world server 150 may display in the virtual environment 102 (e.g. within the virtual sub-environment 120) an identity of each of the one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178, and the corresponding benefit count 142/164 that may be received by the first user 110 when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the selected product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target

real-world data file 138 or a target virtual data file 160 respectively to obtain the selected product 178.

In certain embodiments, based on searching the information relating to data files stored in the memory 156, virtual-world server 150 may determine multiple data files (e.g., multiple real-world data file 138, multiple virtual data files 160 or combinations thereof) that may be used to obtain a same selected product 178. In this case, different ones of the multiple data files 160 may provide different benefit counts 142/164 when used to obtain the same selected product 178. The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) identities of each of the multiple data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178, and a benefit count 142/164 corresponding to each of the data files that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain the selected product 178.

In certain embodiments, when a selected product type 176 is associated with multiple products 178 that are available for selection, virtual-world server 150 may determine a single data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain several products 178 of the same selected product type 176 including the product 178 selected by the first user 110 of the same selected product type 176. In this case, the single data file may provide different benefit counts 142/164 when used to obtain the different products 178 of the selected product type 176. The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) an identity of the single data file (e.g., real-world data file 138 or virtual data file 160) that can be used to obtain each of the multiple products 178 associated with the selected product type 176, one or more of the multiple products 178 (e.g., including the selected product 178) associated with the selected product type 176, and for each indicated product 178, a benefit count 142/164 corresponding to the data file that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain each of the multiple products 178.

In alternative or additional embodiments, in response to receiving a selection of a product type 176 associated with a particular selected experience theme 174, virtual-world server 150 may be configured to search the memory 156 and determine one or more products 178 that are available for selection associated with each selected product type 176. For each product 178 that is available for selection, virtual-world server 150 may be configured to search the information relating to data files (e.g., real-world data file 138 and/or virtual data file 160) stored in the memory 156 to determine one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the product 178 and a benefit count 142/164 corresponding to each determined data file that the first user 110 may receive when

the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may determine different data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain different products 178 belonging to the same product type 176. For example, virtual-world server 150 may determine a first data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain a first product 178 belonging to a product type 176 and determine a second data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain a second product 178 belonging to the same product type 176. The first data file may provide a first benefit count 142/164 when used to obtain the first product 178, and the second data file may provide a second benefit count 142/164 when used to obtain the second product 178.

The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) each of one or more products 178 that is available for selection associated with the selected product type 176, identities of one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain each product 178.

Virtual-world server 150 may be configured to generate a customized virtual experience for the first user 110 based on a selection of one or more products 178 made by the first user 110 in respective product types 176 of an experience theme 174. In one embodiment, the virtual experience simulates an actual experience that the first user 110 may have when using the selected products 178 and thus allows the first user 110 to experience the selected products 178 before obtaining the products 178 by transferring data objects (e.g., real-world data objects 140 or virtual data objects 162) to respective entities that provide the products 178. For example, when the first user 110 selects an experience theme 174 relating to a visual experience in the virtual environment 102 as described above, the first user 110 may select several products 178 associated with multiple product types 176. For example, the first user 110 may select a particular avatar as a selected product 178 from the “avatars” product type 176, one or more buildings selected products 178 from the “buildings” product type 176, and one or more roads as a selected products 178 from the “roads” product type 176. In response to receiving these product selections, virtual-world server 150 may generate a customized virtual sub-environment 120 that simulates a visual experience of the virtual environment 102 using the selected avatar, buildings, and roads. Virtual-world server 150 may allow the first user 110 to navigate the customized virtual sub-environment 120 to experience the simulated visual experience generated for the first user 110. In another example, when the first user 110 selects a travel theme as the selected experience theme 174, the first user 110 may select a particular destination (e.g. city or town) as a selected product 178 from the “destination” product type 176, a particular car as a selected product 178 from the “car” product type 176, a particular hotel as a selected product 178 from the “hotels” product type 176, and a particular excursion as a selected product 178 from the “excursions” product

type 176. In response to receiving these product selections associated with the travel theme, virtual-world server 150 may generate a customized virtual sub-environment 120 that simulates a travel experience of the virtual environment 102 using the selected destination, car, hotel and excursion. Virtual-world server 150 may allow the first user 110 to navigate the customized virtual sub-environment 120 to experience the simulated travel experience generated for the first user 110.

Virtual-world server 150 may be configured to allow the first user 110 (e.g., using first avatar 114) to select one or more data files (e.g. real-world data file 138 and/or virtual data file 160). For example, in response to indicating to the user in the virtual environment 102 (e.g., virtual sub-environment 120) one or more data files (e.g. real-world data file 138 and/or virtual data file 160) that can be used to obtain one or more products 178 associated with an experience theme 174 selected by the first user 110, first user 110 may select a data file (e.g. real-world data file 138 or virtual data file 160). Upon receiving a selection of the data file from the first user 110, virtual-world server 150 may be configured to initiate a registration process to register the first user 110 to obtain the selected data file. In one embodiment, when the data file selected by the first user 110 is a real-world data file 138, the virtual-world server 150 may be configured to transmit a request to the real-world server 130 to take over and register the first user 110 to obtain the real-world data file 138. On the other hand, when the data file selected by the first user 110 is a virtual data file 160, the virtual-world server 150 may be configured to register the first user 110 to obtain the real-world data file 138. As described above, registering the first user 110 may include verifying the identity of the first user 110 and then generating the selected data file (e.g. real-world data file 138 or virtual data file 160) for the first user 110. Once a selected real-world data file 138 or a selected virtual data file 160 has been generated for the first user 110, the first user 110 may store real-world data objects 140 or virtual data objects 162 in the respective real-world data file 138 or a selected virtual data file 160. In one embodiment, the real-world server 130 or the virtual-world server 150 may transfer a default or pre-configured amount of real-world data objects 140 or virtual data objects 162 to the respective real-world data file 138 or a selected virtual data file 160.

Once the first user 110 is registered for the selected data file (real-world data file 138 or virtual data file 160) and the selected data file has been generated and stored at the respective real-world server 130 or virtual-world server 150, first user 110 may use the data file to obtain one or more products 178. For example, when a selected product 178 is a real-world product (e.g., car, train, hotel etc.), the first user 110 may obtain the selected product 178 by commanding the real-world server 130 to transfer a pre-configured amount of real-world data objects 140 from the registered real-world data file 138 of the first user 110 to a target real-world data file 138 of an entity that provides the selected product 178. On the other hand, when a selected product 178 is a virtual-world product (e.g., avatar, virtual building, virtual roads, etc.), the first user 110 may obtain the selected product 178 by commanding the virtual-world server 150 to transfer a pre-configured amount of virtual data objects 162 from the registered virtual data file 160 of the first user 110 to a target virtual data file 160 of an entity that provides the selected product 178 in the virtual environment 102. In response to receiving the command from the first user 110, the respective real-world server 130 or the virtual-world server 150 may transfer the pre-configured amount of data

objects to the target data file, to cause the transfer of the selected product 178 to the first user 110 in the real-world environment or the virtual environment 102 whichever the case may be.

In certain embodiment, before initiating registration of the first user 110 to obtain a selected data file (real-world data file 138 or virtual data file 160), the virtual-world server may be configured to determine whether the first user 110 is eligible to obtain the requested data file. As described above, the virtual-world server 150 may store a user profile 172 associated with the first user 110. Virtual-world server 150 may be configured to determine whether the first user 110 is eligible to obtain the selected data file based on the information stored in the user profile 172 associated with the first user 110. The virtual-world server 150 initiates registration of the first user 110 to obtain the selected data file upon determining that first user 110 is eligible to obtain the selected data file.

In some embodiments, the first user 110 may be registered with a first entity that owns and/or manages the virtual environment 102 but may not be registered with a second entity that provides the data files (real-world data file 138 or virtual data file 160). For example, based on the registration with the first entity, the first user 110 may have generated a user credential 170 that allows the first user 110 to enter the virtual environment 102. The first user 110 may have provided some information (e.g., name, email address, social security number, phone number etc.) when registering with the first entity. This information may be stored as part of the user profile 172 of the first user 110 and associated with the user credential 170 of the first user 110. When the first user 110 selects a data file (real-world data file 138 or virtual data file 160) for registration, the virtual-world server 150 obtains the user profile 172 of the first user 110 associated with the user credential 170 and determines whether the first user is eligible to obtain the selected data file based on information stored in the user profile 172.

FIG. 3 illustrates a flowchart of an example method 300 for performing data interactions in a virtual environment 102, in accordance with one or more embodiments of the present disclosure. Method 300 may be performed by the virtual-world server 150 shown in FIG. 1.

At operation 302, virtual-world server 150 receives a user credential 170 associated with a first user 110, wherein the user credential 170 provides the first user 110 access to a virtual environment 102.

At operation 304, virtual-world server 150 checks whether the user credential 170 is registered at the virtual-world server 150 to provide the first user 110 access to the virtual environment 102. For example, the virtual environment 102 may be owned and/or managed by a first entity. In this case, the first user 110 may need to be registered with the first entity to gain access to the virtual environment 102. If the first user 110 is registered with the first entity, this registration may be stored at the virtual-world server 150 including the user credential 170 that was registered with the first entity to gain access to the virtual environment 102. If the first user 110 is found to be not authorized to access the virtual environment 102 based on the user credential 170, method 300 proceeds to operation 306 where the virtual-world server 150 denies the first user 110 access to the virtual environment 102. On the other hand, if the first user 110 is found to be authorized to access the virtual environment 102, method 300 proceeds to operation 308, where the virtual-world server 150 authorizes, an avatar (e.g., first avatar 114) of the first user 110 to enter the virtual environment 102.

As described above, virtual-world server 150 may be configured to use the user credential 170 collected from the first user 110, generated by the first user 110 or assigned to the first user 110 during real-world data interactions with the first user 110, to verify identity of the first user 110 in the virtual environment 102. Thus, the user credential 170 provides the first user 110 access to the virtual environment 102. For example, the user credential 170 may be used by the virtual-world server 150 to verify that the first avatar 114 belongs to and is controlled by the first user 110.

In one example, user credential 170 may include a username and password generated by the first user 110 as part of registering with the real-world server 130. The virtual-world server 150 may allow the first user 110 to use the same username and password to enter the virtual environment 102 via first avatar 114. In an alternative embodiment, the first user 110 may generate a separate user credential (different from the first user credential 170) as part of registering with the real-world server 130 and use the separate user credential to log into the real-world server 130 and perform real-world data interaction. The first user 110 may separately generate the user credential 170 as part of registering with the virtual-world server 150 (e.g., including registering with a first entity that owns and/or manages the virtual environment 102), for example, during a real-world data interaction with the real-world server 130. Once the first user 110 has registered with the virtual-world server 150 and generated the user credential 170, the first user 110 may use the user credential 170 to log into the virtual-world server 150, gain access to the virtual environment 102 and perform virtual-world data interactions in within the virtual environment 102.

At operation 310, virtual-world server 150 receives a first selection made by the first user 110 of an experience theme 174 using the avatar (e.g., first avatar 114) in the virtual environment 102, wherein the experience theme 174 includes a plurality of selectable product types 176.

As described above, virtual-world server 150 may be configured to facilitate a user (e.g., first user 110) to obtain one or more products 178 in the virtual environment 102 or real-world environment. As shown in FIG. 1, virtual-world server 150 may be configured to store one or more experience themes 174, wherein each experience theme 174 relates to a particular type of experience that can be had by the user 110 in the virtual environment 102 or the real-world environment. An example of an experience theme associated with the virtual environment 102 may include visual elements that make up the virtual environment 102. An example of an experience theme associated with the real-world environment may include a travel experience. An experience theme 174 includes one or more product types 176, wherein each product type 176 is a type of product 178 that relates to the experience theme 174. For example, when the experience theme 174 relates to visual experience of the first user 110 within the virtual environment 102, example product types 176 associated with this experience theme 174 may include, but are not limited to, avatars, buildings, roads, objects, colors and patterns. In another example, when the experience theme 174 relates to a travel experience in the real-world environment, product types 176 associated with this experience theme 174 may include, but are not limited to, cars, buses, trains, travel destinations, hotels, restaurants, and excursions. An experience theme 174 may include several products 178 associated with each product type 176. For example, when the experience theme 174 relates to a visual experience within the virtual environment 102, the “avatars” product type 176 may include several designs of

avatars 114 that may be used as the first avatar 114 of the first user 110 within the virtual environment 102. The “buildings” product type 176 may include a plurality of virtual buildings that can be used to generate the virtual environment 102. The “colors” and “patterns” product type 176 may include a plurality of colors and patterns respectively that can be used to render specific objects or portions of the virtual environment 102. When the experience theme 174 relates to a travel experience in the real-world environment, the “destination” product type 176 may include a selection of different destinations (e.g., cities, towns etc.) the first user 110 may visit, the “cars” product type 176 may include a selection of different cars that be obtained for a travel, the “hotels” product type may include a selection of hotels at the selected destination and the like.

When the first user 110 enters the virtual environment 102 (e.g., using first avatar 114) after logging in to the virtual-world server 150 by providing user credential 170, the first user 110 may navigate to a virtual sub-environment 120 that is configured to facilitate users to obtain products 178 associated with one or more experience themes 174. The virtual sub-environment 120 may be assigned to a particular real-world entity. Once the first user 110 has entered the virtual sub-environment 120, the virtual-world server 150 may present to the first user 110 one or more experience themes 174. For example, the virtual-world server 150 may display the experience themes 174 on a virtual display within the virtual sub-environment 120. Virtual-world server 150 may be configured to allow the first user 110 to select one of the presented experience themes 174. The first user 110 may select one of the experience themes 174 depending on what kind of experience the first user 110 is looking for. For example, the virtual-world server 150 may present a visual experience theme associated with the virtual environment 102 and a travel experience theme associated with the real-world environment. In an example, the first user 110 selects the visual experience theme associated with the virtual environment 102.

At operation 312, virtual-world server 150 receives a second selection made by the first user 110 in the virtual environment 102 of at least one product type 176 from the selected experience theme 174, wherein a first product 178 associated with the selected at least one product type 176 can be obtained by the first user 110 by transferring a first amount of data objects (e.g., real-world data objects 140 and/or virtual data objects 162) to a first entity that provides the first product 178.

As described above, once the virtual-world server 150 receives a selection of an experience theme 174, virtual-world server 150 may be configured to present to the first user 110 within the virtual sub-environment 120, one or more product types 176 associated with the selected experience theme 174. For example, when the first user 110 selects visual experience in the virtual environment 102 as the selected experience theme 174, virtual-world server 150 presents a list of product types 176 associated with the visual experience theme including, but not limited to, avatars, buildings, roads, objects, colors and patterns. Virtual-world server 150 may be configured to allow the first user 110 to select one or more of the presented product types 176 associated with the selected experience theme 174. As the first user 110 selects each product type 176, virtual-world server 150 may be configured to access from the memory 156 one or more products 178 associated with the selected product type 176 and present the one or more products 178 to the first user 110 in the virtual environment 102. For example, when the first user 110 selects the “avatars”

product type 176, virtual-world server 150 may access from the memory 156 one or more avatar designs associated with the “avatars” product type 176 and present the one or more avatar designs to the first user 110 in the virtual-sub environment 120. For example, virtual-world server 150 may display a 2-dimensional or 3-dimensional image of one or more avatars associated with the avatar product type 176. Virtual-world server 150 may allow the first user 110 to select at least one product 178 (e.g., specific avatar design) associated with a selected product type 176. In one embodiment, virtual-world server 150 may allow the first user 110 to select multiple product types 176 related to the selected experience theme 174 and one or more products 178 associated with each of the multiple selected product types 176. For example, the first user 110 may want to build a customized virtual sub-environment 120 and may want to obtain specific visual elements (e.g., avatars, buildings, roads, objects, colors, patterns etc.) to build the virtual sub-environment 120. For this purpose, the first user 110 may select a specific avatar from the “avatar” product type 176, one or more specific buildings from the “buildings” product type 176, one or more roads from the “roads” product type 176, one or more colors from the “colors” product type 176, and one or more patterns from the “patterns” product type 176.

At operation 314, virtual-world server 150 determines a first data file (e.g., real-world data file 138 or virtual data file 160) that provides a first benefit count (e.g., benefit count 142/164) to the first user 110 when the first amount of data objects (e.g., real-world data objects 140 or virtual data objects 162) is transferred from the first data file to a second data file (e.g., another real-world data file 138 or virtual data file 160) associated with the first entity to obtain the first product 178 associated with the selected at least one product type 176.

As described above, once the product selections have been made by the first user 110, virtual-world server 150 may be configured to determine one or more data files (e.g., real-world data files 138 and/or virtual data files 160) that can be used to obtain at least one of the products 178 selected by the first user 110. In this context, virtual-world server 150 may be configured to store information relating to a plurality of real-world data files 138 and virtual data files 160 that may be used by the first user 110 to obtain products 178 associated with experience themes 174. A particular product 178 may be obtained by the first user 110 by transferring a pre-configured amount of real-world data objects 140 from a real-world data file 138 registered to the first user 110 to another real-world data file 138 registered to an entity that provides the particular product 178, and/or by transferring a pre-configured amount of virtual data objects 162 from a virtual data file 160 registered to the first user 110 to another virtual data file 160 registered to the entity that provides the particular product 178. Different products 178 may be obtained by transferring different pre-configured amounts of real-world data objects 140 or virtual data objects 162 to respective data files of entities that provide the products 178. For example, a first product 178 may be obtained by the first user 110 by transferring a first pre-configured amount of real-world data objects 140 or virtual data objects 162, and a second product 178 may be obtained by the first user 110 by transferring a second pre-configured amount of real-world data objects 140 or virtual data objects 162.

In some embodiments, a single data file (e.g., real-world data file 138 or virtual data file 160) may be used to obtain multiple products 178. In some embodiments, any one of multiple data files (e.g., multiple real-world data file 138,

multiple virtual data files 160 or combinations thereof) may be used to obtain a same product 178.

In one or more embodiments, a data file (e.g., real-world data file 138 or virtual data file 160) may provide a benefit count 142/164 to the first user 110, when the first user 110 uses the data file (e.g., real-world data file 138 or virtual data file 160) to obtain a particular product 178 (e.g., by transferring data objects from the data file to a respective data file of an entity that provides the particular product 178). For example, the first user 110 may receive a pre-configured benefit count 142 when the first user 110 uses a particular real-world data file 138 to obtain a particular product 178. Similarly, the first user 110 may receive a pre-configured benefit count 164 when the first user 110 uses a particular virtual data file 160 to obtain a particular product 178. In one embodiment, different data files (e.g., different real-world data files 138 and/or different virtual data files 160) may provide different benefit counts 142/164 to obtain a same product 178. In an additional or alternative embodiment, a single data file (e.g., real-world data file 138 or virtual data file 160) may provide different benefit counts 132/164 to obtain different products 178.

In one or more embodiments, benefit count 142 associated with a real-world data file 138 may be used to obtain one or more products 178 in the real-world environment, and benefit count 164 associated with a virtual data file 160 may be used to obtain one or more products 178 in the virtual environment 102. A higher benefit count 142/164 generally allows the first user 110 to obtain more products 178. Thus, a higher benefit count 142/164 is generally of higher value. The first user 110 may accumulate benefit counts 142/164 received as a result of using a particular data file (e.g., real-world data file 138 or virtual data file 160) to obtain several products 178.

The information stored by the virtual-world server 150 in memory 156 relating to a plurality of real-world data files 138 and virtual data files 160 that may be used to obtain products 178 associated with experience themes 174 may include one or more of an identity of each real-world data file 138 or virtual data file 160, products 178 that may be obtained using the real-world data file 138 or the virtual data file 160, and benefit counts 142 or 164 provided by the real-world data file 138 or virtual data file 160 respectively when used to obtain each of the products 178.

As described above, once the selection of the products 178 associated with a selected experience theme 174 has been made by the first user 110, virtual-world server 150 may be configured to determine one or more data files (e.g., real-world data files 138 or virtual data files 160) that can be used to obtain one or more of the products 178 selected by the first user 110. For a selected product 178, virtual-world server 150 determines one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files that the first user 110 may receive when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the product 178. For example, based on the selected product 178, virtual-world server 150 may be configured to search the memory 156 (e.g., information related to data files stored in the memory 156) for data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178. Based on the search, virtual-world server 150 may determine one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files

that the first user 110 may receive when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the product 178.

At operation 316, virtual-world server 150 indicates to the first user 110 in the virtual environment 102 that the first data file (e.g., real-world data file 138 or virtual data file 160) provides the first benefit count 142/164 to the first user 110 in relation to obtaining the first product 178 associated with the selected at least one product type 176, wherein the indication comprises the first benefit count 142/164.

The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) that the determined one or more data files (e.g., real-world data file 138 and/or virtual data file 160) may provide to the first user 110 the corresponding benefit counts 142/164 when the determined one or more data files are used to obtain the product 178. For example, virtual-world server 150 may display in the virtual environment 102 (e.g. within the virtual sub-environment 120) an identity of each of the one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178, and the corresponding benefit count 142/164 that may be received by the first user 110 when the data file (e.g., real-world data file 138 or virtual data file 160) is used to obtain the selected product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain the selected product 178.

In certain embodiments, based on searching the information relating to data files stored in the memory 156, virtual-world server 150 may determine multiple data files (e.g., multiple real-world data file 138, multiple virtual data files 160 or combinations thereof) that may be used to obtain a same selected product 178. In this case, different ones of the multiple data files 160 may provide different benefit counts 142/164 when used to obtain the same selected product 178. The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) identities of each of the multiple data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the selected product 178, and a benefit count 142/164 corresponding to each of the data files that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain the selected product 178.

In certain embodiments, when a selected product type 176 is associated with multiple products 178 that are available for selection, virtual-world server 150 may determine a single data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain several products 178 of the same selected product type 176 including the product 178 selected by the first user 110 of the same selected product type 176. In this case, the single data file may provide different benefit counts 142/164 when used to obtain the different products 178 of the selected product type 176. The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g.,

within the virtual sub-environment 120) an identity of the single data file (e.g., real-world data file 138 or virtual data file 160) that can be used to obtain each of the multiple products 178 associated with the selected product type 176, one or more of the multiple products 178 (e.g., including the selected product 178) associated with the selected product type 176, and for each indicated product 178, a benefit count 142/164 corresponding to the data file that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain each of the multiple products 178.

In alternative or additional embodiments, in response to receiving a selection of a product type 176 associated with a particular selected experience theme 174, virtual-world server 150 may be configured to search the memory 156 and determine one or more products 178 that are available for selection associated with each selected product type 176. For each product 178 that is available for selection, virtual-world server 150 may be configured to search the information relating to data files (e.g., real-world data file 138 and/or virtual data file 160) stored in the memory 156 to determine one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the product 178 and a benefit count 142/164 corresponding to each determined data file that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may determine different data files (e.g., real-world data file 138 and/or virtual data file 160) that may be used to obtain different products 178 belonging to the same product type 176. For example, virtual-world server 150 may determine a first data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain a first product 178 belonging to a product type 176 and determine a second data file (e.g., real-world data file 138 or virtual data file 160) that may be used to obtain a second product 178 belonging to the same product type 176. The first data file may provide a first benefit count 142/164 when used to obtain the first product 178, and the second data file may provide a second benefit count 142/164 when used to obtain the second product 178.

The virtual-world server 150 may be configured to indicate to the first user 110 in the virtual environment 102 (e.g., within the virtual sub-environment 120) each of one or more products 178 that is available for selection associated with the selected product type 176, identities of one or more data files (e.g., real-world data file 138 and/or virtual data file 160) that can be used to obtain the product 178, and a benefit count 142/164 corresponding to each of the one or more data files that the first user 110 may receive when the data file is used to obtain the product 178. In one embodiment, virtual-world server 150 may additionally indicate to the first user 110 in the virtual environment 102 (e.g., virtual sub-environment 120) an amount of real-world data objects 140 or virtual data objects 162 that are to be transferred to a corresponding target real-world data file 138 or a target virtual data file 160 respectively to obtain each product 178.

At operation 318, virtual-world server 150 receives a command from the first user 110 to register the first user 110 to obtain the first data file (e.g., real-world data file 138 or virtual data file 160).

At operation 320, in response to receiving the command, virtual-world server 150 initiates a registration process to

register the first user **110** to obtain the first data file (e.g., real-world data file **138** or virtual data file **160**).

As described above, virtual-world environment **150** may be configured to allow the first user **110** (e.g., using first avatar **114**) to select one or more data files (e.g., real-world data file **138** and/or virtual data file **160**). For example, in response to indicating to the user in the virtual environment **102** (e.g., virtual sub-environment **120**) one or more data files (e.g., real-world data file **138** and/or virtual data file **160**) that can be used to obtain one or more products **178** associated with an experience theme **174** selected by the first user **110**, first user **110** may select a data file (e.g., real-world data file **138** or virtual data file **160**). Upon receiving a selection of the data file from the first user **110**, virtual-world server **150** may be configured to initiate a registration process to register the first user **110** to obtain the selected data file. In one embodiment, when the data file selected by the first user **110** is a real-world data file **138**, the virtual-world server **150** may be configured to transmit a request to the real-world server **130** to take over and register the first user **110** to obtain the real-world data file **138**. On the other hand, when the data file selected by the first user **110** is a virtual data file **160**, the virtual-world server **150** may be configured to register the first user **110** obtain the real-world data file **138**. As described above, registering the first user **110** may include verifying the identity of the first user **110** and then generating the selected data file (e.g., real-world data file **138** or virtual data file **160**) for the first user **110**. Once a selected real-world data file **138** or a selected virtual data file **160** has been generated for the first user **110**, the first user **110** may store real-world data objects **140** or virtual data objects **162** in the respective real-world data file **138** or a selected virtual data file **160**. In one embodiment, the real-world server **130** or the virtual-world server **150** may transfer a default or pre-configured amount of real-world data objects **140** or virtual data objects **162** to the respective real-world data file **138** or a selected virtual data file **160**.

Once the first user **110** is registered for the selected data file (real-world data file **138** or virtual data file **160**) and the selected data file has been generated and stored at the respective real-world server **130** or virtual-world server **150**, first user **110** may use the data file to obtain one or more products **178**. For example, when a selected product **178** is a real-world product (e.g., car, train, hotel etc.), the first user **110** may obtain the selected product **178** by commanding the real-world server **130** to transfer a pre-configured amount of real-world data objects **140** from the registered real-world data file **138** of the first user **110** to a target real-world data file **138** of an entity that provides the selected product **178**. On the other hand, when a selected product **178** is a virtual-world product (e.g., avatar, virtual building, virtual roads, etc.), the first user **110** may obtain the selected product **178** by commanding the virtual-world server **150** to transfer a pre-configured amount of virtual data objects **162** from the registered virtual data file **160** of the first user **110** to a target virtual data file **160** of an entity that provides the selected product **178** in the virtual environment **102**. In response to receiving the command from the first user **110**, the respective real-world server **130** or the virtual-world server **150** may transfer the pre-configured amount of data objects to the target data file, to cause the transfer of the selected product **178** to the first user **110** in the real-world environment or the virtual environment **102** whichever the case may be.

In certain embodiment, before initiating registration of the first user **110** to obtain a selected data file (real-world data file **138** or virtual data file **160**), the virtual-world server may

be configured to determine whether the first user **110** is eligible to obtain the requested data file. As described above, the virtual-world server **150** may store a user profile **172** associated with the first user **110**. Virtual-world server **150** may be configured to determine whether the first user **110** is eligible to obtain the selected data file based on the information stored in the user profile **172** associated with the first user **110**. The virtual-world server **150** initiates registration of the first user **110** to obtain the selected data file upon determining that first user **110** is eligible to obtain the selected data file.

In some embodiments, the first user **110** may be registered with a first entity that owns and/or manages the virtual environment **102** but may not be registered with a second entity that provides the data files (real-world data file **138** or virtual data file **160**). For example, based on the registration with the first entity, the first user **110** may have generated a user credential **170** that allows the first user **110** to enter the virtual environment **102**. The first user **110** may have provided some information (e.g., name, email address, social security number, phone number etc.) when registering with the first entity. This information may be stored as part of the user profile **172** of the first user **110** and associated with the user credential **170** of the first user **110**. When the first user **110** selects a data file (real-world data file **138** or virtual data file **160**) for registration, the virtual-world server **150** obtains the user profile **172** of the first user **110** associated with the user credential **170** and determines whether the first user is eligible to obtain the selected data file based on information stored in the user profile **172**.

In an example banking use case, the disclosed system and methods described in embodiments of the present disclosure provide motivations to a user (e.g., first user **110**) to sign up for a bank account (e.g., a credit card). In this context, the real-world data file **138** may correspond to a real-world bank account of the first user **110** and the real-world data objects **140** may correspond to the real-world funds in the bank account of the first user **110**. Similarly, virtual data file **160** may correspond to a virtual account (e.g., a crypto account) of the first user **110** and the virtual data objects **162** may correspond to digital currency. Different types of real-world data file **138** may include, but are not limited to, credit card account, debit card account, savings account, and checking account. Benefit count **164** may correspond to reward points that may be earned by the first user **110** when a particular bank account (e.g., credit card or crypto account) is used to purchase a product.

While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and altera-

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tions are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

The invention claimed is:

1. A system comprising:

a memory that stores a user credential associated with a first user; and

at least one processor coupled to the memory, and configured to:

receive the user credential associated with the first user, wherein the user credential provides the first user access to a virtual environment;

authorize, based on the user credential, an avatar of the first user to enter the virtual environment;

receive a first selection made by the first user of an experience theme using the avatar in the virtual environment, wherein the experience theme comprises a plurality of selectable product types;

receive a second selection made by the first user in the virtual environment of at least one product type from the selected experience theme, wherein a first product associated with the selected at least one product type can be obtained by the first user by transferring a first amount of data objects to a first entity that provides the first product;

determine a first data file that provides a first benefit count to the first user when the first amount of data objects is transferred from the first data file to a second data file associated with the first entity to obtain the first product associated with the selected at least one product type;

indicate to the first user in the virtual environment that the first data file provides the first benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the first benefit count;

receive a command from the first user to register to obtain the first data file; and

in response to receiving the command, initiate a registration process to register the first user to obtain the first data file.

2. The system of claim 1, wherein the at least one processor is further configured to:

after the user has obtained the first data file, receive a second command from the first user to transfer the first amount of data objects from the first data file to the second data file associated with the entity to obtain the first product; and

in response to receiving the second command, transfer the first amount of data objects from the first data file to the second data file, to cause a transfer of the first product to the first user in a real-world environment.

3. The system of claim 2, wherein the at least one processor is further configured to:

determine whether the first user is registered with a second entity associated with the virtual environment; if the first user is not registered with the second entity, obtain a user profile of the first user associated with the user credential of the first user;

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determine, based on the user profile, whether the first user is eligible to obtain the first data file; and

initiate a registration process to register the first user to obtain the first data file, if the first user is eligible to obtain the first data file.

4. The system of claim 1, wherein the at least one processor is further configured to:

generate a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with one or more products associated with the selected at least one product type.

5. The system of claim 1, wherein:

the selected experience theme comprises a plurality of products associated with the selected at least one product type;

a second product associated with the selected at least one product type can be obtained by the first user by transferring a second amount of data objects to a second entity that provides the second product; and

the at least one processor is further configured to:

determine a third data file that provides a second benefit count to the first user when the second amount of data objects is transferred from the third data file to a fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and

indicate to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

6. The system of claim 5, wherein the at least one processor is further configured to:

determine that the first data file provides a third benefit count to the first user when the second number of data objects is transferred from the first data file to the fourth data file of the second entity to obtain the second product associated with the selected at least one product type, wherein the first benefit count is larger than the third benefit count; and

indicate to the first user in the virtual environment that the first data file provides the third benefit count to the first user in relation to obtaining the second product associated with the selected at least one product type, wherein the indication comprises the third benefit count.

7. The system of claim 1, wherein the at least one processor is further configured to:

determine that a third data file provides a second benefit count to the first user when the first amount of data objects is transferred from the third data file to the second data file associated with the first entity to obtain the first product associated with the selected at least one product type, wherein the first benefit count is larger than the second benefit count; and

indicate to the first user in the virtual environment that the third data file provides the second benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the second benefit count.

8. The system of claim 1, wherein:

the second selection of the at least one product type comprises:

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a third selection of a first product type; and
 a fourth selection of a second product type; and
 the at least one processor is further configured to:
 generate a virtual sub-environment within the virtual envi-
 ronment, wherein the virtual sub-environment provides the
 first user a virtual experience associated with at least one
 product associated with the first product type and at least one
 product associated with the second product type.

9. A method for performing data interactions in a virtual
 environment, the method comprising:

receiving a user credential associated with the first user,
 wherein the user credential provides the first user
 access to a virtual environment;

authorizing, based on the user credential, an avatar of the
 first user to enter the virtual environment;

receiving a first selection made by the first user of an
 experience theme using the avatar in the virtual envi-
 ronment, wherein the experience theme comprises a
 plurality of selectable product types;

receiving a second selection made by the first user in the
 virtual environment of at least one product type from
 the selected experience theme, wherein a first product
 associated with the selected at least one product type
 can be obtained by the first user by transferring a first
 amount of data objects to a first entity that provides the
 first product;

determining a first data file that provides a first benefit
 count to the first user when the first amount of data
 objects is transferred from the first data file to a second
 data file associated with the first entity to obtain the first
 product associated with the selected at least one prod-
 uct type;

indicating to the first user in the virtual environment that
 the first data file provides the first benefit count to the
 first user in relation to obtaining the first product
 associated with the selected at least one product type,
 wherein the indication comprises the first benefit count;

receiving a command from the first user to register to
 obtain the first data file; and

in response to receiving the command, initiating a regis-
 tration process to register the first user to obtain the first
 data file.

10. The method of claim 9, further comprising:

after the user has obtained the first data file, receiving a
 second command from the first user to transfer the first
 amount of data objects from the first data file to the
 second data file associated with the entity to obtain the
 first product; and

in response to receiving the second command, transfer-
 ring the first amount of data objects from the first data
 file to the second data file, to cause a transfer of the first
 product to the first user in a real-world environment.

11. The method of claim 10, further comprising:

determining whether the first user is registered with a
 second entity associated with the virtual environment;
 if the first user is not registered with the second entity,
 obtaining a user profile of the first user associated with
 the user credential of the first user;

determining, based on the user profile, whether the first
 user is eligible to obtain the first data file; and

initiating a registration process to register the first user to
 obtain the first data file, if the first user is eligible to
 obtain the first data file.

12. The method of claim 9, further comprising:

generating a virtual sub-environment within the virtual
 environment, wherein the virtual sub-environment pro-

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vides the first user a virtual experience associated with
 one or more products associated with the selected at
 least one product type.

13. The method of claim 9, wherein:

the selected experience theme comprises a plurality of
 products associated with the selected at least one prod-
 uct type;

a second product associated with the selected at least one
 product type can be obtained by the first user by
 transferring a second amount of data objects to a
 second entity that provides the second product; and

further comprising:

determining a third data file that provides a second
 benefit count to the first user when the second
 amount of data objects is transferred from the third
 data file to a fourth data file of the second entity to
 obtain the second product associated with the
 selected at least one product type, wherein the first
 benefit count is larger than the second benefit count;
 and

indicating to the first user in the virtual environment
 that the third data file provides the second benefit
 count to the first user in relation to obtaining the
 second product associated with the selected at least
 one product type, wherein the indication comprises
 the second benefit count.

14. The method of claim 13, further comprising:

determining that the first data file provides a third benefit
 count to the first user when the second number of data
 objects is transferred from the first data file to the fourth
 data file of the second entity to obtain the second
 product associated with the selected at least one prod-
 uct type, wherein the first benefit count is larger than
 the third benefit count; and

indicating to the first user in the virtual environment that
 the first data file provides the third benefit count to the
 first user in relation to obtaining the second product
 associated with the selected at least one product type,
 wherein the indication comprises the third benefit
 count.

15. The method of claim 9, further comprising:

determining that a third data file provides a second benefit
 count to the first user when the first amount of data
 objects is transferred from the third data file to the
 second data file associated with the first entity to obtain
 the first product associated with the selected at least one
 product type, wherein the first benefit count is larger
 than the second benefit count; and

indicating to the first user in the virtual environment that
 the third data file provides the second benefit count to
 the first user in relation to obtaining the first product
 associated with the selected at least one product type,
 wherein the indication comprises the second benefit
 count.

16. The method of claim 9, wherein:

the second selection of the at least one product type
 comprises:

a third selection of a first product type; and
 a fourth selection of a second product type; and

further comprising:

generating a virtual sub-environment within the virtual
 environment, wherein the virtual sub-environment
 provides the first user a virtual experience associated
 with at least one product associated with the first
 product type and at least one product associated with
 the second product type.

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17. A non-transitory computer-readable medium that stores instructions which when executed by a processor causes the processor to:

- receive a user credential associated with the first user, wherein the user credential provides the first user access to a virtual environment;
- authorize, based on the user credential, an avatar of the first user to enter the virtual environment;
- receive a first selection made by the first user of an experience theme using the avatar in the virtual environment, wherein the experience theme comprises a plurality of selectable product types;
- receive a second selection made by the first user in the virtual environment of at least one product type from the selected experience theme, wherein a first product associated with the selected at least one product type can be obtained by the first user by transferring a first amount of data objects to a first entity that provides the first product;
- determine a first data file that provides a first benefit count to the first user when the first amount of data objects is transferred from the first data file to a second data file associated with the first entity to obtain the first product associated with the selected at least one product type;
- indicate to the first user in the virtual environment that the first data file provides the first benefit count to the first user in relation to obtaining the first product associated with the selected at least one product type, wherein the indication comprises the first benefit count;
- receive a command from the first user to register to obtain the first data file; and
- in response to receiving the command, initiate a registration process to register the first user to obtain the first data file.

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18. The non-transitory computer-readable medium of claim 17, wherein the instructions further cause the processor to:

- after the user has obtained the first data file, receive a second command from the first user to transfer the first amount of data objects from the first data file to the second data file associated with the entity to obtain the first product; and
- in response to receiving the second command, transfer the first amount of data objects from the first data file to the second data file, to cause a transfer of the first product to the first user in a real-world environment.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions further cause the processor to:

- determine whether the first user is registered with a second entity associated with the virtual environment;
- if the first user is not registered with the second entity, obtain a user profile of the first user associated with the user credential of the first user;
- determine, based on the user profile, whether the first user is eligible to obtain the first data file; and
- initiate a registration process to register the first user to obtain the first data file, if the first user is eligible to obtain the first data file.

20. The non-transitory computer-readable medium of claim 17, wherein the instructions further cause the processor to:

- generate a virtual sub-environment within the virtual environment, wherein the virtual sub-environment provides the first user a virtual experience associated with one or more products associated with the selected at least one product type.

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